



Integrating Clinical Pathways into CDSS using Context and Rules – A Case Study in Heart Disease

Wen Yao

College of Information Sciences and Technology
The Pennsylvania State University
University Park, PA 16802 USA
wxy119@psu.edu

Akhil Kumar

Smeal College of Business
The Pennsylvania State University
University Park, PA 16802 USA
akhil@psu.edu

ABSTRACT

In this paper, we propose an ontology-based approach for integrating clinical pathways into Clinical Decision Support Systems (CDSS) recognizing that flexible pathways are critical for the success of CDSS. Our approach is based on a better understanding of clinical context through ontologies, and bringing them to bear in deciding the right rules for a certain activity. We also describe an approach for integrating and instantiating context specific subprocess fragment into a clinical pathway on the fly based on the needs of a specific case. To illustrate the feasibility of our approach, we present a system architecture and prototype implementation, along with a case study for handling patients with heart disease. The role of semantic web technologies in integrating clinical pathways into CDSS is highlighted.

Categories and Subject Descriptors

J.3 [Computer Applications]: Life and Medical Sciences — *medical information systems*; H.4.2 [Information Systems]: Information Systems Applications — *decision support*

General Terms

Design, Management

Keywords

Clinical pathway, CDSS, rules, context, ontologies

1. INTRODUCTION

Workflows play an important role in clinical environments by delineating the steps through which the treatment of a patient progresses. The temporal order and correct coordination of the various steps is clearly important. The workflows in these environments are called *clinical pathways*. These pathways usually follow well-established standards or clinical guidelines in plain text format. However they differ from other workflows

found in business and production environments in that there are frequent deviations and hence need for considerable flexibility. Typically, a medical facility develops clinical pathways from clinical guidelines based on its local resources and setting. Moreover, the pathway is further customized into a treatment scheme to suit an individual patient's needs.

Clinical workflows are highly dynamic, context sensitive, event driven, and knowledge intensive. In this respect, they are quite unique. In general, a patient interacts with a Primary Care Physician's (PCP) office, a pharmacy, labs, and one or more specialists, etc. In such a setting, it is important to maintain coordination and flow of information among these various entities so as to ensure an optimal outcome. Although many research efforts are geared towards establishing international healthcare standards (e.g., HL7 [28]) and a representation for sharable guidelines (e.g., GLIF [30]) for clinical practice, formal models of executable and flexible clinical workflows are very few (e.g., [11, 47]). Thus, there is an urgent need for new modeling techniques for designing and managing such flexible and knowledge intensive workflows.

The execution of a clinical workflow is highly dependent on the existing body of medical knowledge, available resources, and specific patient data. For example, doctors with different skill levels and fields of expertise may offer differing diagnosis to the same patient, thus leading to different treatment plans for this patient. A sudden rise in a patient's blood pressure may require an additional blood test and alter her treatment in the subsequent pathway. Thus, different pathways can arise based on case specifics and the proclivities of attending doctors. It is important to formally model these scenarios. Another motivation is to reduce medical errors, since medical staff make mistakes in prescribing medications, performing procedures, and even making a diagnosis [6, 23, 42]. For example, a doctor might accidentally prescribe an X-ray test for a pregnant patient, or a drug that is not covered by insurance to another patient. Thus, there is a need for Clinical Decision Support Systems (CDSS) and Computer Interpretable Guidelines (CIGs) [40] to assist care professionals with decision making activities.

In this paper, we propose a methodology for designing flexible clinical processes that capture medical knowledge and context in a common framework. Since the clinical workflow should naturally be aligned with clinical guidelines, we must ensure that it is formal and correct so that integrating decision support into this workflow can be helpful. We developed a formal rule and

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.

IHI'12, January 28–30, 2012, Miami, Florida, USA.

Copyright 2012 ACM 978-1-4503-0781-9/12/01...\$10.00.

context taxonomy using ontologies. The rule taxonomy organizes medical rules into a hierarchy while context encompasses aspects of patient, provider, resources, and environment. We will discuss how contexts are captured, described and summarized, and how semantic rules are developed and modeled. Finally, we will use this ontological knowledge framework to design and execute flexible clinical pathways.

The organization of this paper is as follows. Section 2 discusses related work and our objectives in this study. Our approach is presented in Section 3, including a meta-model and system architecture. Next, Section 4 presents the knowledge base that include ontologies and rules. A case study using our approach for handling patients with heart disease is shown in Section 5, followed by a conclusion and plans for future work in Section 6.

2. RELATED WORK AND OBJECTIVES

Clinical Decision Support System (CDSS) is an interactive computer software designed to assist health professionals with decision making tasks. For example, it can prevent adverse drug events at the point of care [42]. Although many methodologies are employed in designing CDSS, including Bayesian networks, neural networks, and genetic algorithms, we focus on rule-based approaches in this study. Rule-based CDSS, derived from expert systems research [6], are knowledge based systems that integrate a medical knowledge base, patient data, and an inference engine to generate case specific advice. An overview of the state of the art of rule-based CDSS is given in [6, 35, 40].

Although CDSS are meant to assist in clinicians' work, still they are reluctant to adopt them because of unnecessary workflow disruptions [21]. The importance of integrating a CDSS with a clinical workflow has been stressed by many other studies as well [14, 21, 44]. Kawamoto et al. [22] noted that CDSS interventions that are presented automatically and fit into a workflow of medical staff are more likely to be adopted, and a CDSS that recommends actions is better than one that only gives an assessment. Another challenge is to ensure the correctness of decision support by following standard guidelines. The objective of a CDSS is to deliver the "*right knowledge to the right people in the right form at the right time*" [39]. Thus, integrating medical guidelines along with clinical data into a CDSS aims to deliver evidence-based recommendations to care professionals at the point of care.

In this section, we discuss relevant studies that have contributed to the design of flexible clinical workflows from many research communities, including the process management, software engineering and medical informatics communities. Our work is motivated by the research gap we identified.

2.1 Clinical Guidelines

The medical informatics community models clinical processes as guidelines (or care plans), recognizing that some important medical activities in clinical practice are not explicitly supported in workflow modeling. Their focus lies on medical decision making by interpreting situations and events on best practice. They also formalize process definition in a way that reflects clinical tasks and constraints as clinicians perceive them.

Medical guidelines were originally in the form of free-format text documents to assist medical decision making during diagnosis,

management and treatment within different areas of healthcare. Recently, various approaches have been proposed to represent Computer-Interpretable Guidelines (CIGs), such as Arden Syntax, Asbru, EON, GLIF, GUIDE, PRESTIGE, PRODIGY, *PROforma*, and SAGE (see [14, 28, 30, 33, 36, 37]). CIGs can produce personalized recommendations during patient encounters and reduce variance in patient treatment. Peleg et al. [37] reviewed six CIG modeling techniques and established a consensus on a set of common components. These approaches represent clinical guidelines as *plans*, whose components include decisions, actions, and their relationships. *Decision steps* are used for conditional and unconditional routing of the flow, while *action steps* are used to specify a set of tasks or a sub-plan to be carried out. A review of systems using CIGs was conducted by Isern and Moreno [18].

There are other approaches that manually transform constructed diagrams into executable workflows. For example, Ongenaie et al. [32] presented a semi-automatic verification and translation framework based on ontologies and semantic rules to convert clinical protocols from XML flow charts into executable workflows, and integrate it into a service-oriented architecture.

2.2 Clinical Workflow Modeling

The business process management community has devoted a lot of effort in designing customized and flexible clinical processes using formal workflow languages (e.g., BPMN and BPEL) that are executable by existing workflow engines. Lenz and Reichert [24] conducted a survey of IT support for healthcare processes and emphasized the importance of flexible workflow design in clinical settings. Such systems have been developed to enable dynamic changes in predefined process models, such as ADEPT_{flex} [38] and AgentWork [27]. Research on context-aware workflow design [1, 4, 16, 45] also belongs to this area, but focuses more on the role of context in binding specific services or subprocesses, whereas medical knowledge is deemphasized. These techniques are useful and highlight the criticality of context in designing a flexible workflow.

The software engineering community has also contributed toward modeling medical processes. In particular, limitations of current workflow languages have been observed [48], and an alternative approach based on Little-JIL language [46] has been proposed. Little JIL is a language that centers on a coordination diagram of the process described by a hierarchical task decomposition. It helps to coordinate agents and their activities, and allows steps to be performed in sequence or parallel. Approaches for formally defining, analyzing, and verifying medical processes using Little JIL are discussed in [8-10, 34]. This stream of research is useful for its focus on improving patient safety and detecting medical errors, but it does not emphasize flexibility or decision support. Mathe et al. [25] developed the Model-Integrated Clinical Information System using model-based design techniques to represent complex clinical workflows in a service-oriented architecture and then translate them into executable constructs such as Web services. The security specifications are enriched, and treatment protocols are transformed into a process modeling language to promote software reuse and maintainability [26].

2.3 Ontology-based Clinical Process Models

Recently, ontologies have been used in clinical workflows to represent medical knowledge [2, 7, 11, 47]. For example, Dang

et al. [11] built an ontological knowledge framework to represent important entities (e.g., roles and resources) in healthcare. Using this framework, they developed a system [12] that enables the creation and execution of personalized medical workflows in the BPEL model. Similar studies include SEMPAT [2] and KON³ [7], but they focus more on rule-based reasoning. Ye et al. [47] proposed a clinical pathway ontology to represent and exchange pathway-related knowledge. They presented clinical pathways as interconnected hierarchical models including a top-level outcome flow and an intervention workflow level along a care timeline. They used semantic rules for modeling temporal knowledge. Although they used a concrete example of a cesarean section surgery to show the applicability of their methodology, a system architecture and implementation are lacking.

2.4 Objectives of this Study

The variety of workflow management tools simplifies the modeling and execution of clinical workflows. The current process modeling languages provide a powerful modeling vocabulary (e.g., events, messages) and workflow patterns (e.g., loops, complex gateways), which are not available in the CIG modeling languages. However, most existing workflow modeling approaches lack integration with medical data and knowledge, and guidelines are hardly modeled or considered in their system design. On the other hand, CIGs (e.g., Arden Syntax and GLIF) all require an execution engine that is not freely available. Complex workflow patterns cannot be easily modeled in this way, and validation (e.g., checking for possible deadlocks) of such workflows is difficult.

A CDSS needs to follow clinical practice guidelines to provide evidence-based recommendations. Thus, identifying clinical workflows aligned with guidelines, modeling them using a formal and flexible approach, and integrating decision support along the care process are critical to the success of CDSS deployment. Our study aims to bridge the gap in contemporary approaches by tightly integrating guideline-based and semantically enhanced clinical workflows into a CDSS.

3. OUR APPROACH

In this paper, we propose an ontology-based approach for formally modeling medical knowledge and providing decision support. In this section, we first describe a meta model that serves as a roadmap for our approach, and then introduce our system architecture for implementation.

3.1 A Meta Model

Figure 1 represents a meta model that describes the key concepts in healthcare as well as their semantic relationships.

Clinical workflows are composed of *activities* performed by *roles*. *Rules* are derived from *clinical guidelines* and uses medical ontologies for clinical reasoning. They are categorized into *rule groups* according to their semantic meaning. A rule group can be associated with one or more workflow activities and it is automatically triggered when the activity node is reached. The data required to evaluate the rule condition is modeled as *context*, which is critical in the clinical practice. The context captures all entities that have an impact on the clinical process, and has subclasses including *low-level* and *high-level* context. Low-level context is explicit and it is directly collected from user inputs or other data sources (e.g., medical devices, environment

sensors), e.g., patient name and body temperature. They are used to deduct high-level, implicit context by rule-based reasoning, e.g., resource availability and patient situation. When the context is relevant to a clinical decision, rules that meet specific conditions are fired to produce *actions*, which can be *reminders*, *alerts*, and *recommendations*. They are tightly integrated with medical activities as a way to intervene with a clinical workflow. They can provide evidence-based recommendations for medical decision making or generate operational semantics for executing a composite activity that is a loosely coupled subprocess. Further, workflow activities in turn produce context.

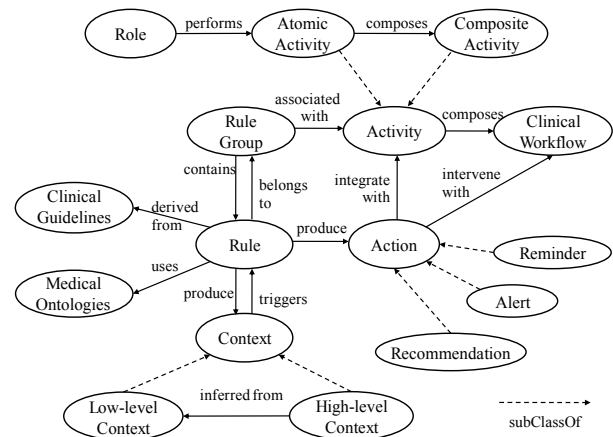


Figure 1. A meta model relating various aspects of a CDSS

3.2 System Architecture

Figure 2 depicts the system architecture for our prototype implementation. Several roles are involved in this system. Knowledge engineers must communicate with domain experts and encode medical knowledge in a computer interpretable format using the ontology and rule editor. *Context ontology* is developed to formally model relevant context variables that have an impact on clinical pathways. Process designers are responsible for creating and maintaining clinical workflow models. The end users include clinicians such as doctors, nurses, and lab staff.

At runtime, a process instance is executed and fed with specific case data (e.g., patient data and clinician data). Clinical context can be acquired from a variety of distributed sources, such as laboratory information systems, surgical information systems, medical sensors, and the workflow engine. Rules are triggered by context if the conditions are met, and produce a series of actions that are processed by the workflow engine. Various participants, i.e., medical staff, manage clinical workflows through web-based management console to perform clinical activities and monitor their current process instances. They can also receive context-aware reminders, alerts, and recommendations in real time.

Thus, this architecture captures the nexus of workflow, rules, and context. It is important to note that a clinician will still be in control. The pathways and alerts generated by the CDSS can be overridden, and they can adjust the alert/reminder settings as per their needs [43]. Their preferences are better integrated into the clinical workflow. We show the open source tools used in our implementation in Figure 2 as well. Next, we discuss three most critical components in more details.

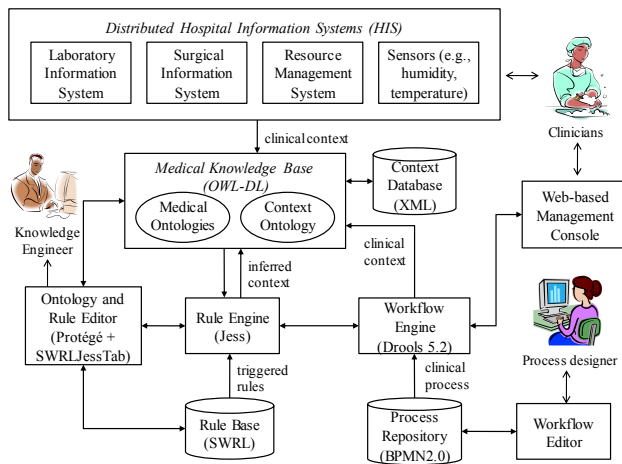


Figure 2. System architecture

Ontology and rule editor: Protégé 3.4 [41] is an open source ontology editor developed by Stanford University. It provides a graphical interface to model ontologies and a plug-and-play environment to allow for flexible prototyping and application development. We used it to encode the context ontology and medical domain ontologies. Further, we encoded the rules for context reasoning in SWRL [17], which is a standard semantic web rule language combining OWL and RuleML. The Protégé plug-in named SWRL Jess tab [29] is used to insert, remove, and edit SWRL rules. Details about the development and modeling of SWRL rules are introduced in Section 4 and 5.

Jess rule engine: The execution of SWRL rules requires the availability of a rule engine to perform rule-based reasoning. There is a variety of open source rule engines, and among them Jess [15] is small, light, and one of the fastest engines available. Jess rule engine is implemented in Java and supports forward and backward chaining. At runtime, context change can trigger relevant rules if requisite conditions are met. The output of firing rules consists of actions such as reminders and alerts that are relevant to the running clinical processes. They are critical in adapting current processes to various situations and supporting medical decision making. Meanwhile, high level contexts may be inferred from low level ones through rule-based reasoning, and they are fed back into the context database for timely updates.

Drools workflow engine: Drools 5.2 [20] serves as the workflow engine that executes predefined clinical processes in BPMN 2.0 [31]. It also maintains the status of all running process instances. The workflow engine continuously produces patient data and updates the context database. On the other hand, it can acquire contextual data and use that to execute a customized subprocess. This is supported by modeling a **composite activity**, using the **ad-hoc subprocess** element in BPMN 2.0. So the execution semantics is obtained at runtime and used to customize an ad-hoc subprocess (see details in Section 4). As the process proceeds, alerts, reminders, and recommendations are provided to the corresponding workflow participants.

3.3 Contributions

Integration of a Clinical Decision Support System (CDSS) with a clinical workflow is a key to its success. Moreover, semantic web technologies can help to create medical ontologies that are

interchangeable and provide rich semantics for customizing a clinical process. The proposed approach is formal, yet also flexible. It is important to note that a doctor will still be in control. The pathways and alerts generated by the CDSS can be overridden, and doctors can adjust the alert settings as per their needs. This paper has the following contributions:

- We propose an ontology model to capture contextual knowledge that impacts the clinical practice. This ontology can be shared and reused among the medical community. It can also be integrated with medical domain ontologies, such as immunization ontology and disease ontology.
- Based on this model, we use semantic rules to encode medical procedural knowledge from clinical guidelines. These rules are used to provide medical decision support through reminders, alerts, and recommendations, and present them to the medical staff at the right time in an unobtrusive manner.
- We model clinical processes using a standard workflow language (BPMN 2.0) and make the dynamic parts adaptable based on the changing context. Thus, each process instance is customized to meet the needs of an individual patient.
- We develop a proof of concept prototype using multiple open source tools and use a case study for illustration.

4. MODELING FLEXIBLE CLINICAL WORKFLOW

In this section, we describe the medical knowledge framework, in particular ontologies and rules. Then, we discuss how it can be used for designing and executing customized clinical processes.

4.1 Clinical Context Model

According to Dey [13], *context is any information that is used to characterize the situation of an entity*. Moreover, a system is context-aware if it uses context to bring to bear in the provision of relevant information and/or services, where *relevance* depends on the user's task. In a clinical workflow, we define context as any information that can be used to influence medical decision making and clinical pathways. For example, a patient's medical history might affect a doctor's prescription (i.e., influence a medical decision); a patient's lab test result will affect her subsequent treatment process (i.e., dynamic routing of pathway). A normal test result will lead to a "follow up check", while an abnormal one may necessitate "sending this patient to hospital."

We classify context into different categories parameterized by a specific set of characteristics and develop a context model based on ontologies since they can facilitate knowledge sharing, logic inference, and knowledge reuse. Ontologies are explicit and formal specifications of the terms in a domain and the relationships among them. They are widely accepted as a useful instrument for modeling context. We used Protégé [41] for representing and visualizing our ontology-based context model. At a preliminary level, we consider four categories of context:

- *Patient-related context* includes medical history, age, gender, physiology data (e.g., blood pressure, body temperature, and heart rate), X-ray test results, mobility state, etc. Patient data can be obtained from human input, medical devices (e.g., physiological monitors), and EMR central repository. It is continuously updated.

- *Clinical staff-related context* concerns information related to doctors, nurses, lab staff, pharmacists, and other providers. They are the major workflow participants in the clinical pathway and their decision can affect the patient treatment. Their attributes include expertise area, level of expertise, gender, availability, workload, desired alert level, etc.
- *Resource-related context* concerns the information about resources (e.g., devices) that are needed for workflow execution. This might include a defibrillator's availability and current location, the number of available wheel chairs, stock levels of medicines, etc. Additional information is required about the environment at a particular facility such as a surgery room's humidity, temperature, and schedule.
- *Location-related context* concerns the current location of persons, medical devices, and other movable assets. This information can be captured by RFID devices, smart phones, or other pervasive computing technologies. This is helpful to deal with emergent situations, such as the need for an OXRB bottle during a surgery.

Figure 3 shows a partial representation of our ontology-based clinical context model that describes the main entities of interest, their properties, and semantic relationships. It shows the contextual information involved in a healthcare process, such as activity, roles, resources, facility, etc. Their relationships are described so as to capture the constraints between these entities, such as the relation “locatedIn” between Person and Location. The pair of relations “treatedBy” and “treats” between Patient and Staff reflect the treatment relationship between them.

The Web Ontology Language (OWL) [5] is a semantic markup language for defining and sharing ontologies. Our context ontology is represented by OWL-DL, which is a subset of OWL with reasonable inference capability. OWL-DL uses an object-oriented approach to describe the structure of a domain in terms of classes and properties. In Figure 3, each context is a *class*, which represents an important entity in the medical domain. For

example, we define classes Patient, Doctor, PulseRate and PhysiologicalData as follows.

```
<owl:Class rdf:ID="Patient"/>
<owl:Class rdf:ID="Doctor"/>
<owl:Class rdf:ID="PulseRate">
  <rdfs:subClassOf>
    <owl:Class rdf:ID="PhysiologicalData"/>
  </rdfs:subClassOf>
</owl:Class>
```

Note that the class PulseRate is defined as a subclass of PhysiologicalData by using rdfs:subClassOf. Class is associated with its attributes through *Datatype property*. Two such properties of the class PulseRate are hasValue and hasTimestamp, presented in OWL-DL as follows:

```
<owl:DatatypeProperty rdf:ID="hasValue">
  <rdfs:domain rdf:resource="#PulseRate"/>
  <rdfs:range rdf:resource="#float"/>
</owl:DatatypeProperty>
<owl:DatatypeProperty rdf:ID="hasTimestamp">
  <rdfs:domain rdf:resource="#PulseRate"/>
  <rdfs:range rdf:resource="#dateTime"/>
</owl:DatatypeProperty>
```

A class can also link with other classes through *Object property*. For example, the class PulseRate is associated with the class Doctor by Object property hasExaminer; another class Patient is associated with PulseRate through the Object property hasPulseRate.

```
<owl:ObjectProperty rdf:ID="hasExaminer">
  <rdfs:range rdf:resource="#Doctor"/>
  <rdfs:domain rdf:resource="#PulseRate"/>
</owl:ObjectProperty>
<owl:ObjectProperty rdf:ID="hasPulseRate">
  <rdfs:domain rdf:resource="#Patient"/>
```

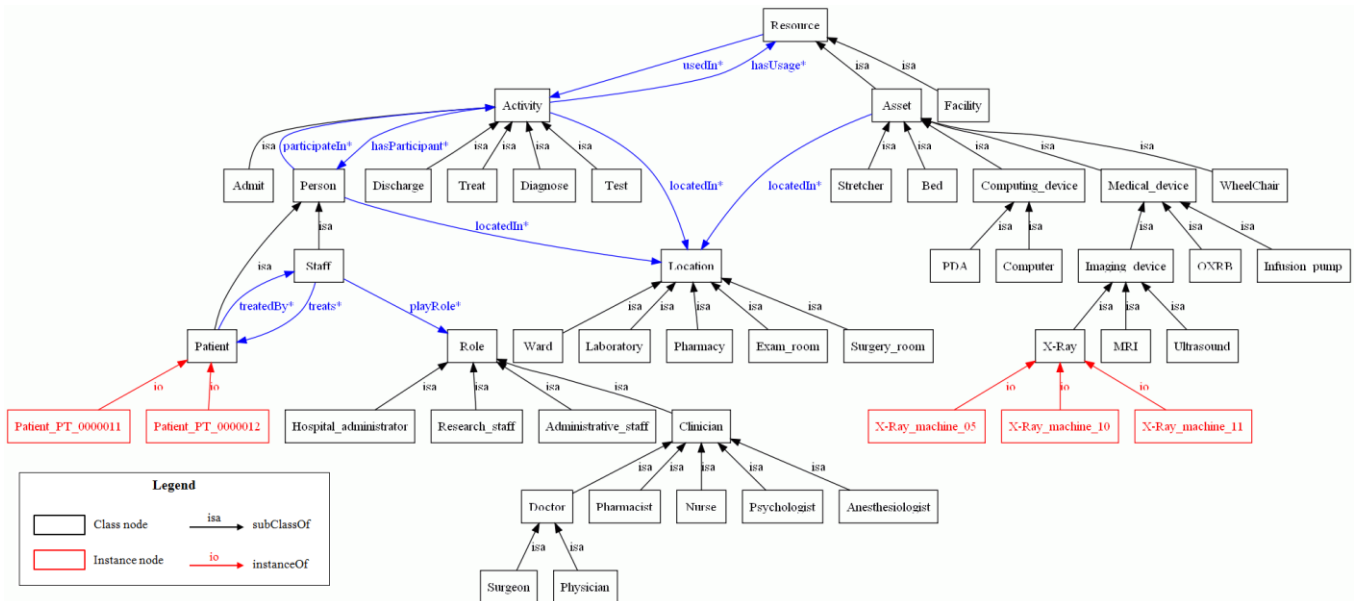


Figure 3. Partial representation of clinical context ontology


```

    <rdfs:range rdf:resource="#PulseRate"/>
</owl:ObjectProperty>

```

Finally, an instance of a class is represented as follows:

```

<PulseRate rdf:ID="PulseRate_10">
  <hasValue rdf:datatype="#float">
    80.0
  </hasValue>
  <hasExaminer>
    <Doctor rdf:ID="Harry"/>
  </hasExaminer>
</PulseRate>

```

Contextual information can be *explicit* or *implicit*. For example, a patient's *age* and *body temperature* are explicit and can be obtained directly, while her *urgency level* is implicit and thus can be inferred from other lower level contexts.

In addition, medical domain ontologies are established to provide a shared understanding of concepts in specialized medical disciplines. They provide important and formal vocabularies for a CDSS. In our prototype, we implemented a heart disease ontology that is illustrated in our case study (see details in Section 5). Other domain ontologies can also be incorporated similarly in future, such as immunization, diabetes, and pediatrics ontologies, to handle other diseases.

4.2 Semantic Rules for Medical Reasoning

Rules in a workflow are important since they embody domain knowledge and are used to make complex business decisions through logical reasoning, often in real time and in response to critical events. They also handle exceptional situations. In a clinical workflow, rules formally represent clinical knowledge that is used to handle complex situations or exceptions. A clinical workflow may have thousands of rules for a variety of purposes but only a small part of the rule set is activated in performing a single workflow step. Thus, rules are categorized

into groups and applied in different stages in a clinical pathway. Managing and integrating medical knowledge in the form of rules into a clinical workflow is critical to the outcome of decision support.

We use Semantic Web Rule Language (SWRL) [17] to encode rules for user-defined reasoning. SWRL is the standard rule language for the semantic web based on OWL and RuleML. It utilizes the typical logic expression “antecedent $\rightarrow$ consequent” to represent semantic rules. Both antecedent (rule body) and consequent (rule head) can be conjunctions of one or more atoms written as “atom₁ $\wedge$ atom₂... $\wedge$ atom_n”. If all the atoms in the antecedent are true, then the consequent must also be true. In our examples in Section 5, the symbol $\wedge$ means conjunction, ?x stands for a variable (i.e., an instance or individual), and $\rightarrow$ means implication. The consequent part of a triggered rule can be used to update context database, issue reminders and alerts, and customize an ad-hoc subprocess (i.e., composite activity).

Similar rules are placed in the same category called a *rule group*. Rules in the same group usually serve the same purpose (e.g., diagnosis or treatment), and are triggered simultaneously during process execution. Figure 4 presents a semantic hierarchy of rules. It was developed based on the consultation with relevant clinical websites and documents such as AHRQ guidelines. This hierarchy serves as an ontology for organizing rules into meaningful groups and subgroups. For example, the rule group “prescription checking” is further subdivided into subgroups that involve “allergy checking”, “drug interactions checking”, “dose checking”, and “insurance checking”. In case a drug is not covered by insurance, the system can propose an alternative drug that may be covered to the physician for approval. For now, we only have a limited number of rules focusing on heart diseases. Example rules in this respect are described in our case study in Section 5. We plan to collaborate with health professionals in specialized disciplines and enrich the rule hierarchy.

We first express rules in plain English **If-Then** format and then

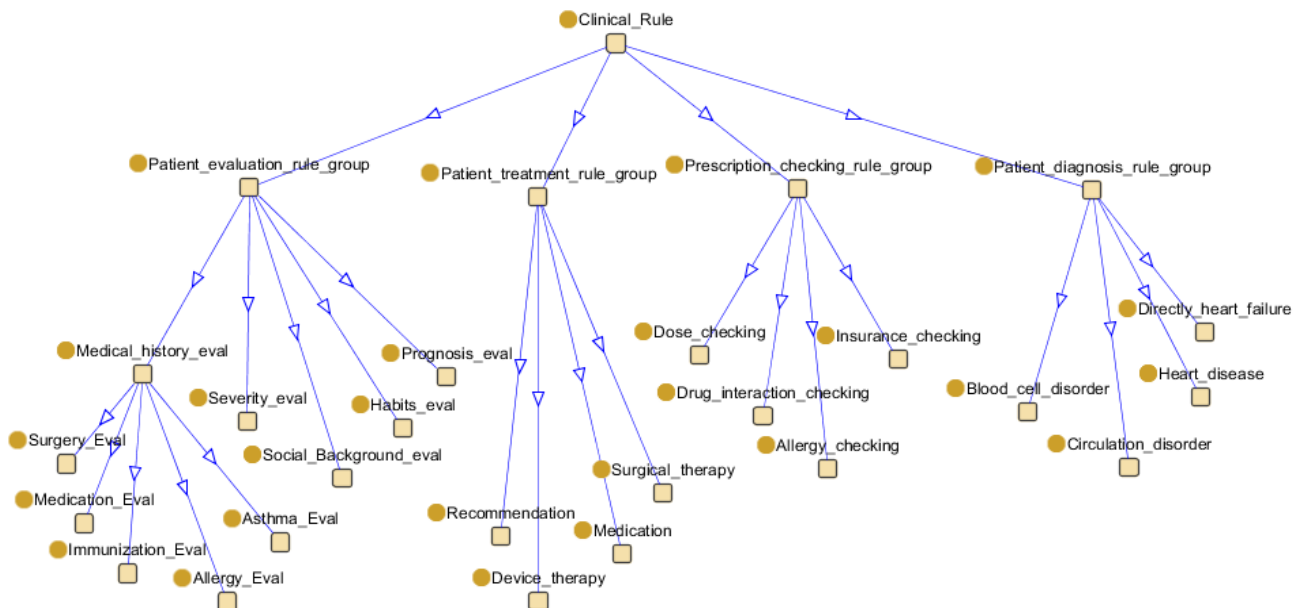


Figure 4. A semantic hierarchy of rules based on clinical knowledge

convert them into SWRL language. SWRL rules are embedded in our system in the rule repository as shown in Figure 2. They are stored as OWL individuals with their associated knowledge base. SWRL reasoning with the Jess rule engine is conducted in the following procedure. First, SWRL rules are translated into Jess rules using SWRLJessBridge [29] and added to the Jess rule engine; then, the ontologies and the knowledge base are translated into Jess facts and introduced into the rule engine as well; third, the Jess rule engine performs the reasoning and produces results in the Jess format; and finally, these results are translated back into the OWL format. Rules can be executed recursively since the knowledge base is continuously updated by newly inserted facts. The reasoning results are tightly integrated with clinical pathways and provide evidence-based decision support for clinicians. Next, we will discuss how semantic rules are used to design flexible clinical pathways.

4.3 Integration of Rules in Clinical Pathways

The key innovation in this work is the tight integration of pathways and rules to improve the operation of CDSS systems, while *recognizing that at any stage in a clinical workflow the subsequent path is determined by the context*. Hence, flexibility is of the essence. The decision on the next path to take can be determined by applying rules. Moreover, the context does not only come from the current case data, but is also based on the specific pathway that was taken to reach this point in the workflow. It may as well involve other aspects like resource availability, clinician status and environment conditions.

4.3.1. Activity + Context $\rightarrow$ Rules

A *composite activity* is broadly or loosely defined, and it can be composed of *atomic activities* that are finely grained and more concrete. The semantics for such activity composition is based on the actual context data. For example, “prescription check” is a composite activity. In reality, it may involve one or more other activities such as “allergy check”, “drug conflict check”, etc. depending upon the medical condition of a specific patient.

There is an m-to-n relationship between (activity, context) on the one hand and rules on the other hand. This means that a certain contextual situation with regards to an activity may need application of many rules, while a given rule may apply to many (activity, context) situations. Thus a context where a patient does not have any allergies does not require an allergy check, and similarly if a patient does not carry insurance, then insurance checking is omitted. These relationships are expressed as “(Activity, Context) $\rightarrow$ Rules”:

- (Medication, allergic) $\rightarrow$ allergy check rule(s)
- (Admission, broken leg) $\rightarrow$ wheel chair check rule(s)
- (Admission, emergency) $\rightarrow$ emergency procedure rule(s)
- (Do tests, X-ray) $\rightarrow$ dosage check
- (Revaluation, overweight) $\rightarrow$ diet recommendation rule(s)

Hence, during the prescription activity, if a patient is allergic to some drugs, an allergy check is needed. At admission time, if there is an emergency then a different admission procedure that deviates from the normal one is used. But clearly, at admission time we do not need to check whether the patient has an allergy or not. Thus, only relevant rules are triggered for a specific activity, based on the actual context.

4.3.2. BPMN 2.0 for Modeling Healthcare Processes

BPMN 2.0 [31] can support various types of activities and workflow patterns. It is executable and supported by the Drools-Flow 5.2 workflow engine [20] (also known as jBPM5). We use a variety of BPMN 2.0 elements to model flexible clinical pathways. An atomic activity is indivisible and used to compose processes and composite activities. It has various types. For example, a *human task* (or user task) requires human interaction with the assistance of a software application while a *manual task* is performed by a human. A *service task* is done automatically by a Web service or an automated application. A *script task* is executed by the workflow engine, and a *rule task* usually involves some rules to handle complicated business logic. These task types are used to support the modeling of different kinds of medical activities.

An ad-hoc subprocess is a composite activity that serves as a container for storing other activities without knowing their execution semantics at design time. Figure 5 shows the XML schema of the BPMN 2.0 element **AdHocSubProcess**. We can see that there are no explicit control flow dependencies among the containing activities. Instead, a more concrete description is realized at runtime based on the context data. For example, “Do tests” activity is an ad-hoc subprocess that contains a number of other activities, such as “blood test”, “X-ray”, “MRI”, etc. *It is not possible to predict what tests the doctor will prescribe until the patient is examined*. Thus the *execution semantics* of “Do tests” is triggered when a patient’s medical condition is diagnosed.

```
<xsd:element name="adHocSubProcess"
type="tAdHocSubProcess" substitutionGroup="flowElement"/>
<xsd:complexType name="tAdHocSubProcess">
  <xsd:complexContent>
    <xsd:extension base="tSubProcess">
      <xsd:sequence>
        <xsd:element name="completionCondition" type=
          "tExpression" minOccurs="0" maxOccurs="1"/>
      </xsd:sequence>
      <xsd:attribute name="cancelRemainingInstances"
        type="xsd:boolean" default="true"/>
      <xsd:attribute name="ordering"
        type="tAdHocOrdering" default="Parallel"/>
    </xsd:extension>
  </xsd:complexContent>
</xsd:complexType>

<xsd:simpleType name="tAdHocOrdering">
  <xsd:restriction base="xsd:string">
    <xsd:enumeration value="Parallel"/>
    <xsd:enumeration value="Sequential"/>
  </xsd:restriction>
</xsd:simpleType>
```

Figure 5. XML schema of AdHocSubProcess in BPMN 2.0

We use Drools-Expert [19] as a complementary tool to realize the *execution semantics* of an ad-hoc subprocess in the BPMN process model. Each ad-hoc subprocess is associated with a number of Drools rerouting rules with the same ID. Once the subprocess is reached, these rules are automatically triggered and produce the execution semantics for the activities in that ad-hoc

subprocess. This subprocess is completed when there is no active instance, specified by the attribute **completionCondition** in Figure 5. Then the workflow engine will execute the subsequent activities. Thus, different contexts can trigger rules and produce different subprocess fragments on the fly. Drools-Expert works with the following mechanism. It is trained about processes by inserting the current state of the processes as part of the working memory data. On the other hand, the clinical context data that is relevant to the current process instance is loaded from our OWL medical knowledge base, transformed into a Java class using the adapter, and then inserted into the working memory as facts. Drools-Flow listens for any event generated from Drools-Expert and executes the activities specified in the event.

5. CASE STUDY – Heart Failure Process

In this section, we will walk through a simplified case of patient heart failure and discuss how it can be handled using our approach. Further, we show the results from our prototype.

Managing the flow of heart failure patients is a difficult task because this process involves a lot of medical knowledge, and the concrete activities for lab tests and treatments can vary a lot depending on a patient's condition. In this case study, we adopted the heart failure ontology from the HEARTFAID team [3] and integrated it into our ontological knowledge base. Semantic rules are developed to reason patient situations in the context of heart failure (e.g., diagnosis of systolic or hypertensive heart failure). The rules are based on the knowledge from American Academy of Family Physicians (AAFP) (<http://www.aafp.org/>) and American Heart Association (<http://www.heart.org>).

Patient Evaluation Rules (PER): These rules are responsible for the evaluation of a patient's medical condition, physiological data, medication, etc. Thus, they should be triggered during *physical examination*. For example, the following rule detects that a patient is suspected of *chronic heart failure*:

PER1: Patient(?p) $\wedge$ hasSymptoms(?p, Dyspnea) $\wedge$ hasSymptoms(?p, Fatigue) $\wedge$ hasSymptoms(?p, Peripheral_edema) $\wedge$ hasSymptoms(?p, Palpitations) $\wedge$ hasSymptoms(?p, Coughing) $\wedge$ hasSymptoms(?p, Nausea) $\wedge$ hasSymptoms(?p, Neurologic_deficit) $\rightarrow$ hypotheticalDiagnosis(?p, Chronic_heart_failure)

Patient Diagnosis Rules (PDR): These rules support diagnosis of a patient after her symptoms and signs are obtained. They should

be triggered during *patient diagnosis*. For example, the following rule determines that the patient may have *systolic heart failure*:

PDR1: Patient(?p) $\wedge$ hypotheticalDiagnosis(?p, Chronic_heart_failure) $\wedge$ hasTestResults(?p, Chest_X-ray_abnormal) $\wedge$ hasTestResults(?p, ECG_abnormal) $\wedge$ hasTestResults(?p, BNP_value_higher_than_100_pg_per_ml) $\wedge$ hasTestResults(?p, LVEF_lower_than_40_percent) $\rightarrow$ suggestedDiagnosis(?p, Systolic_heart_failure)

Patient Treatment Rules (PTR): These rules are concerned with how to treat a patient based on her diagnosis results. They should be triggered during *patient treatment*. The following two rules suggest treatment for patients detected with systolic heart failure, depending on their symptoms:

PTR1: Patient(?p) $\wedge$ suggestedDiagnosis(?p, Systolic_heart_failure) $\rightarrow$ suggestedTreatment(?p, ACE_inhibitor) $\wedge$ suggestedTreatment(?p, Beta_blocker)

PTR2: Patient(?p) $\wedge$ suggestedDiagnosis(?p, Systolic_heart_failure) $\wedge$ hasSymptoms(?p, edema) $\wedge$ hasSymptoms(?p, congestion) $\rightarrow$ suggestedTreatment(?p, ACE_inhibitor) $\wedge$ suggestedTreatment(?p, Beta_blocker) $\wedge$ suggestedTreatment(?p, Diuretic)

Prescription Checking Rules (PCR): These rules check the correctness of prescriptions. Thus, they should be triggered during the *prescription* activity. The following two rules detect patient-drug interaction and drug-drug interaction respectively.

PCR1: Patient (?p) $\wedge$ allergicTo (?p, Aspirin) $\wedge$ Prescription (?Rx) $\wedge$ hasPrescription (?p, ?Rx) swrlb:contains (?Rx, Aspirin) $\rightarrow$ hasAlert (?p, This patient is allergic to Aspirin)

PCR2: Patient (?p) $\wedge$ hasPrescription (?p, ?Rx1) $\wedge$ Prescription (?Rx1) $\wedge$ Prescription (?Rx2) $\wedge$ interactWith (?Rx1, ?Rx2) $\rightarrow$ forbidRx (?p, ?Rx2)

The workflow process for handling a patient with heart disease is represented in Figure 6. This process is modeled in BPMN 2.0 in Drools-flow. There are *four ad-hoc subprocesses* in this workflow model: *Do tests-ASP1*, *Treatment-ASP2*, *Medication-ASP3*, and *Therapy-ASP4*, where *Treatment-ASP2* is a nested one. Figure 7 shows the process log generated by our prototype

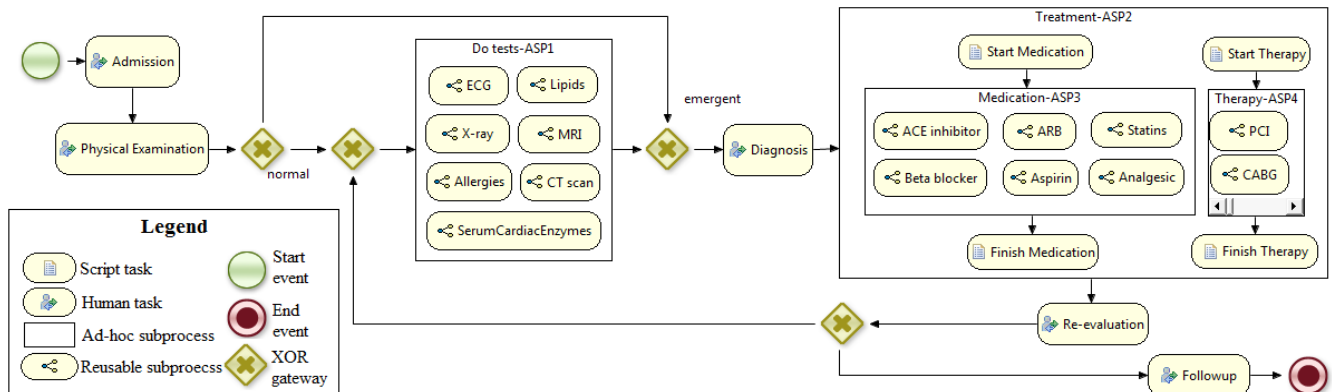


Figure 6. A loosely coupled workflow model for handling patients with heart disease

when the workflow model in Figure 6 is instantiated for a specific patient, who needs thrombus breaking, is not intolerant of ACE inhibitor, and shows signs of high LDL, etc (i.e., contextual data). It reflects the execution semantics for the ASP “Do-tests-ASP1” and “Treatment-ASP2” (shown in the red box) for this specific patient. More complicated workflow patterns can also be modeled and instantiated similarly. Such process logs are usually hidden from the end user but they can be analyzed further for studying medical information of heart failure patients along their pathways using process mining.

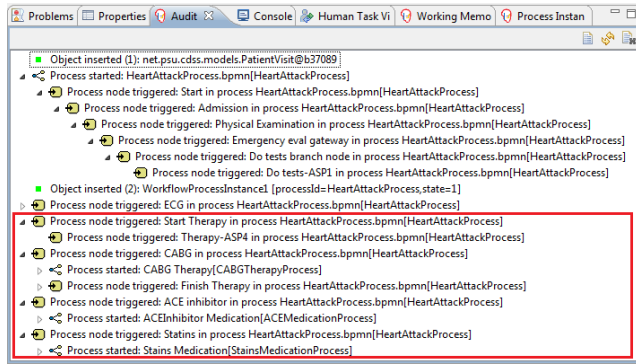


Figure 7. Screenshot of the process log for a specific patient

From this brief case study, we can see that our clinical processes are well structured in general, yet some complex medical activities are loosely-defined at design time using the ad hoc subprocess construct. As a result, we only need to configure the dynamic parts at runtime without starting from scratch for the entire process. Further, we argue that a BPMN 2.0 model is more understandable by care professionals since it provides a graphical representation (as shown in Figure 6). It also has rich semantics for modeling medical tasks and thus is suitable in the healthcare context. Although the current system focuses on dynamic routing of clinical pathways according to rule-based reasoning, we will extend this system by developing more scenarios to demonstrate medical decisions support for care professionals, e.g., how to deal with comorbidity problems and potential counter-indicator of drugs. This system can be first deployed in a small clinic and then in small or mid-sized hospitals.

A key objective in developing and implementing such a system is to improve quality as reflected in concrete measures. Although detailed discussion of quality is beyond the scope of the current work, some key measures of quality (KPI's) for our purposes are: *number of treatment errors because of drug interactions (or allergies), number of diagnosis errors, number of cases of treatment not covered by patient's insurance, number of treatment failures for lack of available resources, complication rate per patient, patient satisfaction, etc.* These metrics will inform the evaluation of the impact of the proposed system. One way to use these metrics is by means of a post hoc audit. For example, we can compare the number of treatment errors from drug interactions after the CDSS system is in place versus the corresponding number before it is installed by examining the patient log over two different periods of equal lengths.

6. CONCLUSIONS

The goal of our work is to study new ways for designing CDSS by integrating rules and context tightly into a workflow model. In

this paper, we propose to use semantic web technologies such as ontologies and semantic rules to design and execute adaptable clinical pathways, taking into account medical knowledge and detailed contextual data to support medical decision making.

Although we model medical knowledge with SWRL rules here, we further plan to develop a translation mechanism to make the knowledge convertible between Arden syntax and SWRL to make our approach more acceptable in the medical community. Further, we plan to develop a user friendly interface to present alerts and reminders to the medical staff, who should be given the right to override suggested guidelines. Such violations will be logged in the system as well. Improved system usability is critical for increasing the acceptance and adoption of such a CDSS system in the clinical settings. In the next phase, we aim to test the current prototype in a practical environment using quality metrics. An audit of past errors without a CDSS and also under CDSS operation can reveal the improvement realized from the new system. Other issues of interest are how to create a contextual summary automatically for a doctor based on a patient's record, and how to allow a doctor to customize her alert settings so the alerts are useful but not distracting.

Thus, the proposed research poses several technical challenges, and promises many benefits to the healthcare community at a time when costs are running out of control. It can lead to better, “smarter” routing of clinical workflows by applying rules with a deeper understanding of context. Moreover, various kinds of alerts can improve patient safety, and reduce treatment errors. Finally, better and quicker recommendations can be generated at various decision points.

7. ACKNOWLEDGMENTS

This work was supported in part by the Center for Integrated Healthcare Delivery Systems (CIHDS) and by the Smeal College of Business at Penn State.

8. REFERENCES

- [1] Adams, M., et al. Extensible and context-aware exception handling for workflows. In *Proceedings of CoopIS'07*, 2007, 95-112.
- [2] Alexandrou, D., et al. SEMPAT: semantic adaptive and personalized clinical pathways. In *2009 International Conference on eHealth, Telemedicine, and Social Medicine (eTELEMED '09)*, 2009, 36-41.
- [3] Anonymous. HF Ontology, University of Calabria, Department of Electronics, informatics, Systems (DEIS), Italy, <http://lis.irb.hr/heartfaid/ontology/>, 2008.
- [4] Ardissono, L., et al. Context-aware workflow management. L. Baresi, P.F., G.-J. Houben ed. *ICWE*, 2007, 47-52.
- [5] Bechhofer, S., et al. OWL web ontology language reference. In *W3C recommendation*, 10. 2006–2001.
- [6] Berner, E.S. Clinical decision support systems: State of the art. Technical Report. *AHRQ Publication No. 09-0069-EF*, Rockville, Maryland, 2009.
- [7] Ceccarellia, M., et al. A guideline engine for knowledge management in Clinical Decision Support Systems (CDSSs) *Proceedings of SEKE*, 2009, 252-257.
- [8] Chen, B., et al., Analyzing medical processes. In *Proceedings of the 30th international conference on Software engineering*, (Leipzig, Germany, 2008), 623-632.

- [9] Christov, S., et al., Rigorously defining and analyzing medical processes: An experience report. In *MoDELS Workshops*, (Nashville, TN, USA, 2008), 118–131.
- [10] Clarke, L.A., et al. Using software engineering technology to improve the quality of medical processes. In *Companion of the 30th international conference on Software engineering*, Leipzig, Germany, 2008, 889–898.
- [11] Dang, J., et al. An ontological knowledge framework for adaptive medical workflow. *Journal of Biomedical Informatics*, 41 (5). 829–836.
- [12] Dang, J., et al. Personalized medical workflow through semantic business process management. In *11th International Conference on Enterprise Information Systems (ICEIS)*, 2009.
- [13] Dey, A.K. Understanding and using context. *Personal Ubiquitous Comput.*, 5 (1). 4–7.
- [14] Fieschi, M., et al. Medical decision support systems: old dilemmas and new paradigms. *Methods Inf Med*, 42 (3). 190–198.
- [15] Friedman-Hill, E. Jess, the rule engine for the java platform. *Sandia National Laboratories*. <http://www.jessrules.com/>, 2005.
- [16] Heravizadeh, M. and Edmond, D. Making workflows context-aware: a way to support knowledge-intensive tasks *Proceedings of the fifth on Asia-Pacific conference on conceptual modelling*, Wollongong, Australia, 2008, 79–88.
- [17] Horrocks, I., et al. SWRL: A semantic web rule language combining OWL and RuleML. *W3C Member submission*, 21.
- [18] Isern, D. and Moreno, A. Computer-based execution of clinical guidelines: A review. *International Journal of Medical Informatics*, 77 (12). 787–808.
- [19] Jboss Community. Drools Expert. <http://www.jboss.org/drools/drools-expert>, 2011.
- [20] Jboss Community. Drools Flow. <http://www.jboss.org/drools/drools-flow.html>, 2011.
- [21] Karsh, B.T. Clinical practice improvement and redesign: how change in workflow can be supported by clinical decision support *AHRQ Publication No. 09-0054-EF*, Rockville, Maryland, 2009.
- [22] Kawamoto, K., et al. Improving clinical practice using clinical decision support systems: a systematic review of trials to identify features critical to success. *British Medical Journal*, 330 (7494).
- [23] Kohn, L.T., et al. To err is human: building a safer health system. *National Academy Press, Washington DC*.
- [24] Lenz, R. and Reichert, M. IT support for healthcare processes-premises, challenges, perspectives. *Data & Knowledge Engineering*, 61 (1). 39–58.
- [25] Mathe, J., et al. Model-based design of clinical information systems. *Methods Inf Med*, 47 (5). 399–408.
- [26] Mathe, J.L., et al. A Model-Integrated, Guideline-Driven, Clinical Decision-Support System. *IEEE Software*, 2009, 54–61.
- [27] Müller, R., et al. Agentwork: a workflow system supporting rule-based workflow adaptation. *Data & Knowledge Engineering*, 51 (2). 223–256.
- [28] Nakamura, M., et al. A multi-layered architecture for process variation management. In *World Conference on Services - II (SERVICES-2 '09)*, 2009, 71 - 78.
- [29] O'Connor, M.J., et al., Supporting rule system interoperability on the semantic web using SWRL and Jess. In *Fourth International Semantic Web Conference (ISWC2005)*, (Galway, Ireland, 2005), 974–986.
- [30] Ohno-Machado, L.a.G., J.H. and Murphy, S.N. and Jain, N.L. and Tu, S.W. and Oliver, D.E. and Pattison-Gordon, E. and Greenes, R.A. and Shortliffe, E.H., Barnett, G. The Guideline Interchange Format (GLIF). *Journal of the American Medical Informatics Association*, 5 (4). 357.
- [31] OMG. Business Process Model and Notation (BPMN) Version 2.0, <http://www.omg.org/spec/BPMN/2.0/>, 2011.
- [32] Ongena, F., et al. Towards computerizing intensive care sedation guidelines: design of a rule-based architecture for automated execution of clinical guidelines. *BMC Medical Informatics and Decision Making*, 10 (3).
- [33] OpenClinical Arden Syntax. http://www.openclinical.org/gmm_ardensyntax.html.
- [34] Osterweil, L.J., et al., Engineering Medical Processes to Improve Their Safety. In *IFIP International Federation for Information Processing*, (Geneva, Switzerland, 2007), Boston Springer, 267–282.
- [35] Peleg, M. and Tu, S. Decision support, knowledge representation and management in medicine. *Yearbook of medical informatics*. 72–80.
- [36] Peleg, M., et al. Comparing models of data and knowledge for guideline-based decision support. *Report SMI-2002-0923*, 2002.
- [37] Peleg, M., et al. Comparing computer-interpretable guideline models: A case-study approach. *Journal of the American Medical Informatics Association*, 10 (1). 52–68.
- [38] Reichert, M. and Dadam, P. ADEPT flex—supporting dynamic changes of workflows without losing control. *Journal of Intelligent Information Systems*, 10 (2). 93–129.
- [39] Schreiber, G., et al. Knowledge Engineering and Management: The Common KADS Methodology. *The MIT Press*, Cambridge, MA, 2000.
- [40] Sittig, D.F., et al. Grand challenges in clinical decision support. *Journal of Biomedical Informatics*, 41 (2). 387–392.
- [41] Standford. Protégé: an open source ontology editor and knowledge-based framework. <http://protege.stanford.edu/>.
- [42] Teich, J.M., et al. Clinical decision support in electronic prescribing: recommendations and an action plan. *British Medical Journal*, 12 (4). 365.
- [43] van der Sijs, H., et al. Overriding of drug safety alerts in computerized physician order entry. *Journal of the American Medical Informatics Association*, 13 (2). 138–147.
- [44] Wears, R.L. and Berg, M. Computer technology and clinical work: still waiting for Godot. *The Journal of the American Medical Association (JAMA)*, 293 (10). 1261–1263.
- [45] Wieland, M., et al. Towards context-aware workflows. In *Proceedings of the Workshops and Doctoral Consortium of CAiSE07*, 2007.
- [46] Wise, A. Little-JIL 1.0 Language Report. Technical report (UM-CS-1998-024), Department of Computer Science, University of Massachusetts, Amherst, MA 1998.
- [47] Ye, Y., et al. An ontology-based hierarchical semantic modeling approach to clinical pathway workflows. *Computers in Biology and Medicine*, 39 (8). 722–732.
- [48] Zhu, L., et al. Challenges observed in the definition of reference business processes. In *BPM 2007 Workshops*, 2007, 95–107.